COMBINATORICS ON ONE PAGE

Simplexes in Pascal, and the trees on their edges

Diagonals of the triangle number the faces, and Cayley's n^{n-2} counts the trees grown leg by leg.

1 / THE RULE AND THE PICTURE

Row n of Pascal's triangle is the face count of the simplex on n vertices

$n=0$	1							
$n=1$	1	1						
$n=2$	1	2	1					
$n=3$	1	3	3	1				
$n=4$	1	4	6	4	1			
$n=5$	1	5	10	10	5	1		
$n=6$	1	6	15	20	15	6	1	
$n=7$	1	7	21	35	35	21	7	1
$n=8$	1	8	28	56	70	56	28	8	1

ones
integers
triangular
tetrahedral
pentatope

vertices n	V	E	triangles	tetra.	5-cells
3: triangle	3	3	1		
4: tetrahedron	4	6	4	1	
5: 4-simplex	5	10	10	5	1
6: 5-simplex	6	15	20	15	6
7: 6-simplex	7	21	35	35	21

A face is a choice of vertices, so the j-faces number $\binom{n}{j+1}$:

$$\#\{j\text{-faces}\} = \binom{n}{j+1}$$

2 / FACES AND EULER'S FORMULA / ALTERNATING ALONG A ROW

$V - E + F - \cdots$ is 1 for the solid simplex and $1 + (-1)^n$ for its surface

Row n alternates to zero, $1 - n + \binom{n}{2} - \cdots = 0$, so the faces of the filled simplex alternate to 1, and removing the single top cell leaves the boundary with $\chi = 1 + (-1)^n$. The tetrahedron's surface gives $4 - 6 + 4 = 2$, your $V - E + F = 2$, and the 5-simplex's gives $6 - 15 + 20 - 15 + 6 = 2$.

$$\chi(\partial\Delta) = 1 + (-1)^n : \quad n = 3,4,5,6,7 \longmapsto 0,\ 2,\ 0,\ 2,\ 0$$

3 / TREES ON n LABELLED POINTS / CAYLEY'S COUNT

$1, 3, 16, 125, 1296, \ldots = n^{n-2}$, found by counting each shape's labellings



$n = 4$: $12 + 4 = 16$ $\qquad$ $n = 5$: $60 + 60 + 5 = 125$

n	2	3	4	5	6
trees n^{n-2}	1	3	16	125	1296

A shape contributes $n!/|\mathrm{Aut}|$ labelled trees: the 4-path $4!/2 = 12$, the star 4; the 5-path 60, the fork 60, the star 5.

4 / WHERE THE TWO MEET / A TREE IS $n-1$ OF THE SIMPLEX'S EDGES

The $\binom{n}{2}$ edges of the simplex offer $\binom{\binom{n}{2}}{n-1}$ choices, and n^{n-2} of them are trees



n	3	4	5	6	7
edges $\binom{n}{2}$	3	6	10	15	21
choices $\binom{\binom{n}{2}}{n-1}$	3	20	210	3003	54 264
trees n^{n-2}	3	16	125	1296	16 807
not trees (a cycle)	0	4	85	1707	37 457

For $n = 4$ the 20 in the middle of Pascal's row 6 is 16 trees plus 4 triangles with an isolated vertex; for $n = 5$ the 85 non-trees are 15 four-cycles and 70 triangles plus an edge.

5 / BUILDING UP LEGS / PASCAL'S ROW INSIDE CAYLEY'S COUNT

Fix the vertex n and count the trees in which it has d legs

$n\backslash d$	1	2	3	4	5
4	9	6	1		
5	64	48	12	1	
6	625	500	150	20	1

A new vertex joins a tree as a leg on an existing vertex; in the Prüfer code of length $n-2$ the label n appears $d-1$ times when it has d legs, giving $\binom{n-2}{d-1}(n-1)^{n-1-d}$ trees, and summing over d is the binomial theorem on row $n-2$:

$$\sum_{d=1}^{n-1} \binom{n-2}{d-1}(n-1)^{n-1-d} = \big((n-1)+1\big)^{n-2} = n^{n-2}$$

ASIDE / THE MATRIX-TREE THEOREM

The Laplacian of K_n has eigenvalues 0 and n ($n-1$ times), and the product of the non-zero ones over n is n^{n-2}.

ASIDE / CHECK IN PYTHON

`round(np.linalg.det((n*np.eye(n)-1)[1:,1:]))` gives 16 for $n = 4$; `comb(comb(n,2),n-1)` gives 20.

Convention. n counts vertices, so the simplex is $(n-1)$-dimensional and its edge graph is K_n; trees are labelled, as in the problem of joining n points by $n-1$ intervals.

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